CH UDAYA KUMAR

Bengaluru, Karnataka, India | +91 73381 77048 | udayreddy1400@gmail.com
linkedin.com/in/uday554 | github.com/UKR555

PROFESSIONAL SUMMARY

Computer Science & Engineering graduate specializing in AI/ML & Backend Engineering with hands-on internship experience at ParkNSecure. Proficient in Java, Spring Boot, Python, YOLOv8, and RESTful APIs with strong foundations in DSA and System Design. Seeking SDE / AI Engineer roles.

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL, JavaScript, HTML5, CSS3
- **AI & Machine Learning:** Artificial Intelligence (AI), Machine Learning (ML), YOLOv8, YOLOv8-Pose, GRU Neural Networks, OpenCV, EasyOCR, Tesseract OCR, Weka API, NumPy
- **Backend & Web:** Spring Boot 2.7, RESTful APIs, Spring Data JPA, Hibernate
- **Databases:** MySQL, MongoDB
- **Tools & Frameworks:** Git, GitHub, Docker, Postman, IntelliJ IDEA, VS Code, Maven, Figma, Vercel
- **Core CS Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming, Operating Systems, Computer Networks, System Design

PROFESSIONAL EXPERIENCE

ParkNSecure Private Limited — Bengaluru, India
AI / ML Intern — May 2025 – June 2025

- Architected an **Automatic Number Plate Recognition (ANPR)** and vehicle monitoring module for an AI-driven smart parking management system.
- Fine-tuned custom YOLOv8 detection models for vehicle localization and license plate bounding-box extraction, routing cropped OpenCV frames into a Tesseract OCR engine for text parsing.
- Integrated real-time video stream ingestion with Spring Boot backend services, optimizing frame processing workflows and reducing manual monitoring overhead.

PROJECTS

AI Malpractice & Cheating Detection System **[GitHub Repo]**
Technologies: Python, Django, YOLOv8-Pose, OpenCV, Three.js, Chart.js

- Architected a multi-modal computer vision monitoring system leveraging YOLOv8 for unauthorized object detection (mobile phones) and YOLOv8-Pose for 17-point skeletal pose estimation.
- Formulated spatial-temporal tracking logic in OpenCV to detect posture anomalies, seat abandonment, out-of-frame student movements, and unauthorized document passing in real time.
- Developed an interactive web dashboard for real-time video stream processing, visual telemetry animations, and real-time Chart.js metric logging.

Grocery Customer Churn Prediction Engine **[GitHub Repo]**
Technologies: Java 11, Spring Boot 2.7, Weka 3.8 ML Library, Bootstrap 5

- Architected a predictive analytics Java web application utilizing Spring Boot 2.7 and the Weka 3.8 ML library to evaluate customer shopping behavioral patterns and demographics.
- Benchmark-tested and integrated 5 machine learning classifiers (Decision Trees J48, Random Forest, Naive Bayes, Logistic Regression, and SVM/SMO) with automated runtime selection of the highest-performing model based on cross-validation metrics.
- Engineered an interactive web portal featuring real-time churn probability scoring, risk tier categorization, and automated personalized customer retention recommendations.

E-Commerce Backend Platform
Technologies: Java, Spring Boot, MySQL, Spring Data JPA, REST APIs, Postman

- Architected a production-ready monolithic backend service featuring 10+ RESTful API endpoints for user authentication, product catalog routing, and order lifecycle management.
- Formulated a relational database schema using MySQL, establishing clean entity mappings with Spring Data JPA and building paginated query filters.
- Enforced strict 3-tier MVC architecture (Controller, Service, Repository layers) and integrated global exception handling to guarantee 99.9% application stability.

EDUCATION

Gandhi Institute of Technology and Management (GITAM) — 2022 – 2026
Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML) — **CGPA: 7.91 / 10.0**

V.K.J PU College — 2020 – 2022
Pre-University Education (PCMB) — **Grade: 77.83%**

Vijaya Marie School — 2019 – 2020
Secondary School Certificate (Class X) — **Grade: 89.76%**

CERTIFICATIONS & ACHIEVEMENTS

- **MongoDB Administration Certification (Infosys Springboard):** Certified in document schema design, aggregation frameworks, indexing strategies, and CRUD operations.
- **Smart India Hackathon (SIH):** Selected for university hackathon team 'HARBINGERS'; engineered backend API contract rules and database integration layers.

ADDITIONAL INFORMATION

- **Spoken Languages:** English (Professional), Kannada (Native), Telugu (Native), Hindi (Conversational)
- **Technical Interests:** Advanced Data Structures, Open-Source Contributions, AI Systems Architecture